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Student Smart Card

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***Abstract*— Radio Frequency Identification (RFID) is used in many applications such as public transport ticketing, campus access, etc. It allows us quick and legit authentication of a person owning RFID tag. In this paper, we are proposing a system that will use RFID tag as an ID for students which will be used for day to day tasks like marking attendance, taking books from library or as a hall ticket in exams. Also, it will be used for payments provided in college including stationary, canteen and college bus. To identify a student uniquely, this system will use it's fingerprint to generate a key which will be use to access content of RFID tag. Another part of this project is Analytics. The data collected through different use of RFID tag by student will be stored in cloud (server). This data will be analyzed to provide insights in data regarding performance of students and use of different services provided by college. It can be used identify different issues regarding student's performance and hence to design and devise solution to overcome it.**

Keywords— analytics, data management, rfid, student data

I. Introduction

For a long there has been use of different registers/diaries to store student data. That mainly includes attendance, library records, marks and curriculum activities. With all this data produced, these systems simply turn out to be data logging system where the data is only inserted. There is no easy way to figure out any results to drive a decision based on these data, unless it is pre-organized for that specific purpose.

This approach is partially automated with introduction of cloud as a service to store and retrieve data of students. Yet, the procedure involved mainly in inserting data is too. manual which sometimes makes it slow. But still, running some scripts over this data leaves some scope for analytics.

Thus, manual traditional ways cannot completely handle the amount of data produced by a college efficiently and there is some need of automation. Also, analyzing this large data using computers can provide insights to students' performance. It can also provide cause of different issues and thus help us devise a possible strategy to solve it. This will help college managements to make decisions, ultimately improving students' performance.

Use of RFID tags provides an easy identifier for students. Data stored in it will be easily and securely read by RFID scanners. This data will be send to remote cloud server which where it will stored. This stored data will be processed and will provide real time analytics and results about working of college.

II. Related Work

Existing systems require manual operations to carry out day to day task in college. These systems are slow and tedious to use when it comes to large amount of data. Following are some examples of existing system.

A. Traditional System

Traditional System uses different diaries and registers to store data. So, a college has to maintain lot of diaries to keep this data. Different procedures/operations done in these systems are slow and tedious. Updating any piece of existing data into these registers can be a difficult job and leaves prone to some human errors. Searching any specific detail is tedious. Also, analysis of this data is very difficult and has very limited scope. But, if the arrangement of data is pre-planned for acquiring some results, like total count of students in a lecture, it can be done but requires manual counting.

B. Cloud Based System

Another upgrade added to traditional system, in recent years, is use of cloud [1]. Here, some IT organizations provide cloud based web application for management of student data. Cloud storage, not completely, yet replaces/eliminates use of registers mentioned in traditional systems as the existing data on temporary created sheet is manually inserted through web application. Searching and retrieving data becomes handy and faster. Updating a record also becomes easy. They allow anytime, anywhere access, while providing security to student data.

The drawback in cloud based systems comes into picture in manual insertion and updation of data. It requires a lot of human intervention.

Let's consider a scenario for marking attendance. First the teacher has to pass a sheet of paper around class making them to sign in front of their roll number. Then, when lecture's over, the teacher will go back to his/her desk. Switch on her computer and open the web application to mark the attendance of that lecture, immediately. Or there can be a second case in which, that teacher may prefer to mark it on weekend of all lectures conducted in that week. The drawback here (for

second case) is the system fails to keep up with real time data. That is, it goes out of synchronization with real time data. So, making the inconsistency where the system may send defaulter emails to a student.

That is not expected, thus produces a big flaw.

C. RFID based system

In [2], a RFID based system is introduced that uses fixed placed scanner connected to computer and all data goes to database in computer. It presents a system that uses RFID technology to enroll and take attendance of students. This system tends to waste some time of students who would require standing in a queue to enroll their attendance. Also, accurate attendance may not be taken as proxy attendance is always a problem in educational institute. Thus, a method is required to authentic the user of RFID tag.

III. Proposed System

The proposed system uses RFID tag to uniquely identify a user and his/her fingerprint template to generate a key to access the content of RFID. This data is sent to cloud server through Wi-Fi (internet) where it is stored, processed and is accessible to students and staff. The diagram below shows basic architecture of system.

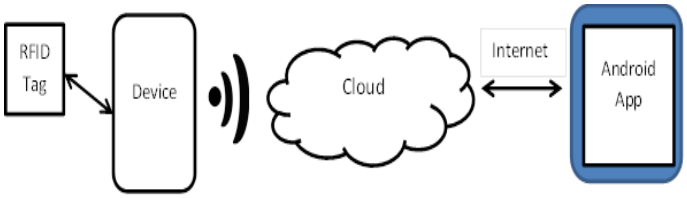

Fig. 1. Basic Architecture of Student Smart Card

A. Components and their functionalites

1) Device & RFID Tag: RFID Tag will work as unique identifier for student. It will store student data like name, class, unique id, ePay id, library books id subscribed, etc. We will be using MIFARE 1K RFID Tags. They have 1KB of EEPROM storage. This 1KB is organized in 16 sectors of 4 blocks each. (One block consists of 16 byte). Each sector is protected by 2 6 byte keys. So, we can securely store data related to different application in different sectors.

Device consists of AtMega1284 at its core connected to different hardware modules through different interfaces. The device will read/write RFID Tags through RFID Scanner module. Log all those transaction in EEPROM of MCU. Upload/download data to/from cloud servers. The detailed description of hardware components used for device is given in next section.

Device will have following functionalities:

- Read/Write to RFID Tags.
- Log every transaction.
- Store temporary student data from cloud server for attendance and other activities.
- Upload transaction log to cloud server.
- Generate a key from fingerprint of user to access RFID Tag contents.

2) Cloud Server: Cloud Servers are computers where student data will be stored. They may be in-house or in remote location. An application running on these computers will be responsible for authentication and reception, transmission, security of stored data on cloud. This application will also provide different techniques to analyse stored student data.

Cloud Server will provide following functionalities:

- Store student data securely.
- Control access to data.
- Provide different ways for analyzing stored data.
- Provide access to data to students and staff through client application.
- Provide authentication mechanisms to device before uploading/downloading data.

3) Client Application: Client Application will be an Android App and a website. These will provide access for staff and students to their data. Also, these application will send request to view visualizations of analysis done on their data. Students will be able to ask queries to their teachers through it.

Client Application will provide following functionalities:

- Provide access to student to his/her attendance, marks, ePay account, library details, personal detail.
- Receive notifications regarding low attendance, end of library subscription period, exam timetable, etc.
- Allow access to analytics data.

B. Hardware Components used in device.

Below is architecture of device:

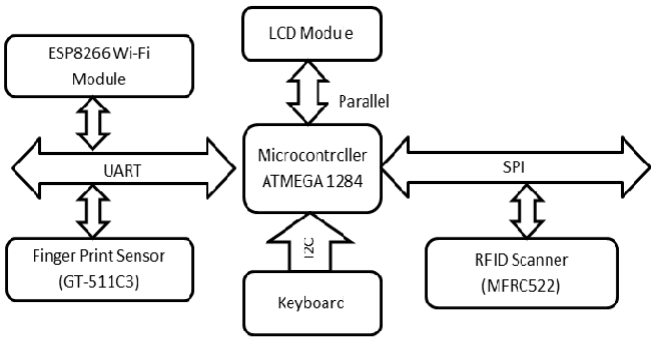

Fig. 2. Architecture of device

Following is detailed description of hardware components used for device:

1) Microchip AtMega1284 microcontroller: The Microchip ATmega1284 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC

architecture. By executing powerful instructions in a single clock cycle, the ATmega1284 achieves throughputs close to 1MIPS per MHz.

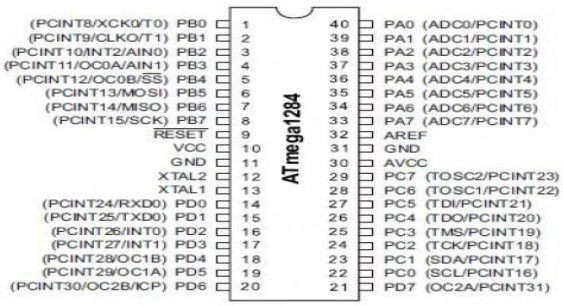

Fig. 3. Microchip AtMega 1284 Pinout

The ATmega1284 provides the following features: 128Kbytes of In-System Programmable Flash Program memory, 4096bytes EEPROM, 16384bytes SRAM, 32 general purpose I/O lines, 32 general purpose working registers, three flexible Timer/Counters with compare modes, internal and external interrupts, a serial programmable USART, a byte oriented Two-wire Serial Interface, an 8 channel, 10-bit ADC with optional differential input stage, a programmable Watchdog Timer with Internal Oscillator and an SPI serial port.

2) MFRC-522 RFID Scanner: The MFRC522 is a highly integrated reader/writer IC for contactless communication at 13.56 MHz. The MFRC522 reader supports ISO/IEC 14443 A/MIFARE and NTAG. The MFRC522s internal transmitter is able to drive a reader/writer antenna designed to communicate with ISO/IEC 14443 A/MIFARE cards and transponders without additional active circuitry.

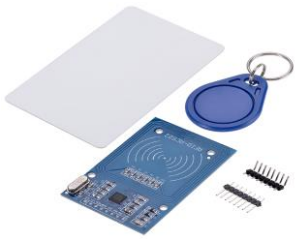

Fig. 4. MFRC-522 RFID Scanner Module

The receiver module provides a robust and efficient implementation for demodulating and decoding signals from ISO/IEC 14443 A/MIFARE compatible cards and transponders. The digital module manages the complete ISO/IEC 14443 A framing and error detection (parity and CRC functionality. It uses SPI interface to communicate with MCU.

3) ESP 8266-01 Wifi Module: ESP8266 is a complete and self-contained Wi-Fi network solutions that can carry software applications. It has built-in TCP/IP protocol stack. Supporting UART interface, it can provide wifi access to microcontorller based application.

Fig. 5. ESP8266-01 Wifi Module

It can work in both P2P or soft-AP modes. There is a set of AT commands that allow communication with it for MCU. Built-in cache memory will help improve system performance and reduce memory requirements.

4) LCD Module: LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. LCDs are economical; easily programmable; have no limitation of displaying special and even custom characters (unlike in seven segments), animations and so on. 16x2 LCD means it can display 16 characters per line and there are 2 such lines.

Fig. 6. LCD Module

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

5) GT-511C3 Fingerprint Sensor:

Fig. 7. Fingerprint Sensor

The GT-511C3 FPS (fingerprint scanner) is a small embedded module that consists of an optical sensor mounted on a small circuit board. The optical sensor scans a fingerprint and the microcontroller and firmware provides the modules functionality which automatically processes the scanned fingerprint. The interface of the FPS is very basic consisting of only four pins VCC5V, ground, serial transmit and serial receive.

C. Procedures/Operations in Student Smart Card.

Following are some procedures through which 'Student Smart Card' will be used.

1) Marking Attendance: Pre-requisites are every student should have unique id stored on his/her RFID registered to system. Teacher's device has access to class he/she is about to lecture.

a) Teacher switches on device and connects to Wi-Fi Network.

b) Selects the class he/she about to take lecture.

c) The relative data (including Name, RFID stud_id, key) is downloaded and stored to EEPROM in device.

d) Teacher may disconnect to Wi-Fi and goes to lecture.

e) Teacher passes his/her device to class.

f) Student takes device place his/her RFID near RFID scanner and finger to finger print sensor.

g) The device scans both, uses finger print to generate a key to access RFID stud_id and logs attendance to EEPROM.

h) The student passes device to next student.

i) After lecture, teacher goes back to Wi-Fi zone and uploads the data to Cloud.

j) The data is stored in Cloud and can be retrieved through app.

2) Library Subscription: Pre-requisites are every book in library is assigned with RFID with a book_id. Students RFID card has two lib_id s (provided student can subscribe up to 2 books at a time).

a) Librarian has a device. Student takes a book.

b) Librarian scans books' RFID.

c) Librarian takes students RFID (with Fingerprint) and scans it.

d) Device checks library sector in RFID if there is any book already allocated to both the lib_id.

e) If there is any slot empty, go to step f. Else warns librarian on display and book is not allocated.

f) Provided a slot is empty, device writes to student's RFID library sector.

g) Librarian scans book RFID to allow device to write that book is subscribed by lib_id.

h) This transaction is logged in EEPROM and is uploaded to Cloud.

i) Librarian may choose to delete the log later. The purpose of log is just to store till it is successfully committed to Cloud.

3) Using ePay (payments at canteen or stationary):

Students can use their RFID to pay for services provided in college like canteen, stationary and college bus. The person receiving payment is called merchant in below text.

Pre-requisites are merchant has a device which is connected to Cloud through Wi-Fi. Student RFID card has enough credits to pay.

a) Merchant enters amount to device to be paid.

b) Student brings RFID card near to scanner (with Fingerprint).

c) Device scans card's ePay sector for ePay_id and sends the request to cloud.

d) Cloud verifies if user's card has enough of credit to complete the transaction and replies back to cloud.

e) If Cloud replies positive, device completes the transaction and logs it. Else, device reports merchant on display and terminates transaction.

f) Device sends transaction completion to Cloud and is committed to Cloud.

IV. Result Analysis

Like in [3], data collected through different activities (in college) will be analyzed to acquire some new insights and some predictions can be made that would help staff and management to take decision beneficial for students. For example, we can use regression model to predict if student can be defaulter (attendance below 75%), so we can take action to avoid it. Below is a demonstration for analysis of library records.

A. Dataset Description.

The dataset is randomly script generated. It includes 200 library records for 23 students with 14 books in library. It has 4 columns namely, Name, Book, Days and Month where Name is student's name, Book is name of book he/she subscribed, Days is number of days the book was subscribed and Month is month in which book was subscribed.

B. Analysis of Library DataSet

Library records can be used to extract different interests of different students in different classes. Now-a-days, a college library has book library plus digital library for online publications. In this dataset, we have only considered book library records.

Plot below show many times a book was subscribed

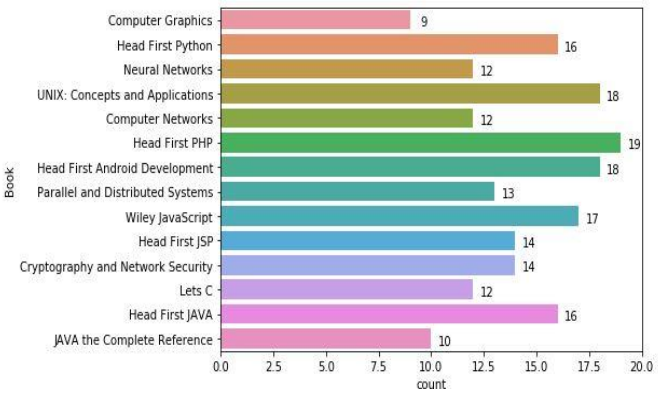

Fig. 8. Number of times a book subscribed

The plot can be used to identify books that are regularly subscribed. It can help management to find what topics are of students' interest. And, to take decisions to improve by conducting events like special lectures to promote this interest.

Plot below shows distribution of number of days value month-wise.

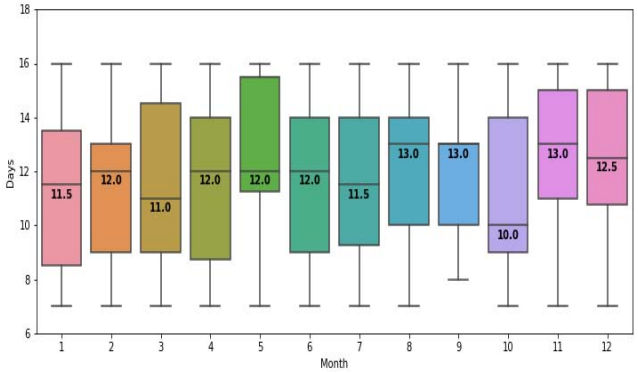

Fig. 9. Days value distribution month-wise

The plot can help to understand the reasons for subscription of book (interest or exams). It is preferred to take median values than mean to avoid noise from some outliers.

Plot below describes how many records of individual records exist in record.

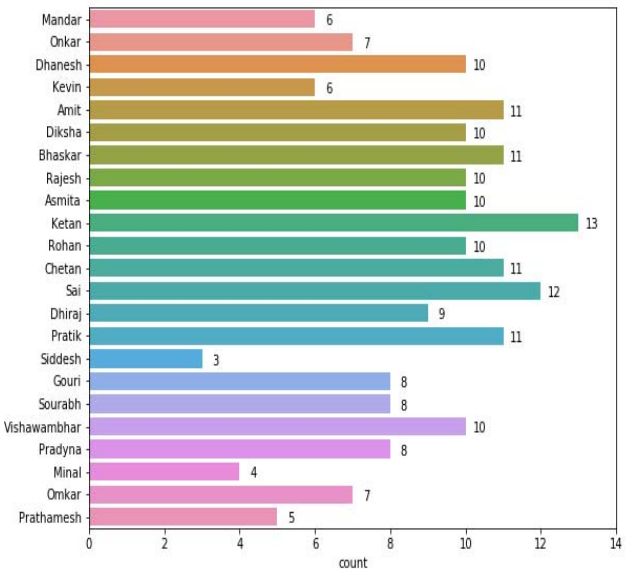

Fig. 10. Number of records of student

The plot can be helpful to identify the regular and non-regular member to library which can helpful to understand the reasons for difference between frequencies of their visits.

Let us take example of Ketan as his number of records is high for student specific analysis. Ketan has subscribed 'Head First Python', 'Wiley JavaScript', 'Head First Android Development', 'Cryptography and Network Security', 'JAVA the Complete Reference', 'Computer Networks', 'Parallel and Distributed Systems', 'Head First JAVA' and 'Head First PHP' books.

Plot below describes how many times Ketan subscribed each book.

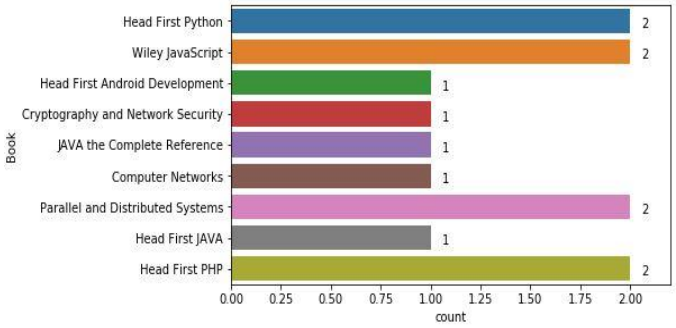

Fig. 11. Number of subscription done by Ketan

This can help analyze what books a student (Ketan) prefers and what can be his subjects/topics of his interest.

Plot below describes on what months Ketan had how many subscriptions.

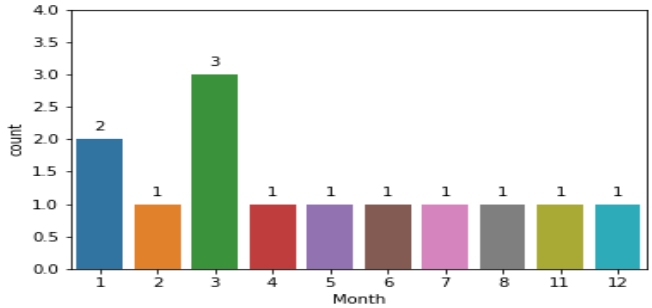

Fig. 12. Month-wise subscription of Ketan

This can help staff understand whether students prefer going library on exams or during normal academics. The books subscribed during normal academics may provide some insights of student's interest. And thus, college management can provide different additional courses in field of students interests so they can excel and be more employable.

V. Conclusion

We have created a system for college and other institutes to automate their day to day activities. This will help normal functioning of colleges more efficient. Also, leveraging analytics over data collected through different activities in college would help management to understand the needs and interests of students leading to data driven decisions.

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